Chapter 18: Electric Charge & Electric Field

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PHY201

Lecture 18

Part I

Electric Charge

Outline of Part I: Electric Charge

Outline of Part I: Electric Charge

The Electromagnetic Force

Why Study Electrostatics?

- The **electromagnetic force** is one of the four fundamental forces of nature.
- It is responsible for *all* macroscopic contact forces: friction, tension, normal force, adhesion, cohesion.
- **Electrostatics**: the study of electric phenomena due to charges that are (at least temporarily) stationary.

Examples of Electrostatic Phenomena

- Lightning
- Static cling after removing clothes from a dryer
- A charged balloon sticking to a wall
- Dust attracted to a freshly polished car
- Hair standing on end after removing a wool hat

Outline of Part I: Electric Charge

Electric Charge

Definition: Electric Charge

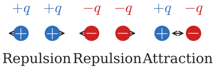
Electric charge is a physical property of matter that causes it to experience a force when near other charged matter. Only two types of charge exist in nature: **positive** and **negative**.

Charge Carriers

- **Proton**: positive charge
- **Electron**: negative charge
- **Neutron**: no charge

Fundamental Rule

- **Like charges repel.**
- **Unlike charges attract.**
- Force decreases with the square of distance.



The SI Unit of Charge: the Coulomb

Elementary Charge

The **elementary charge** is the magnitude of charge on a single proton or electron:

$$|q_{e,p}| = 1.60 \times 10^{-19}\,\mathrm{C}$$

The SI unit of charge is the **coulomb** (C).

- All charges in nature are **integer multiples** of the elementary charge.
- $q_p = +1.60 \times 10^{-19}\,\mathrm{C}; \qquad q_e = -1.60 \times 10^{-19}\,\mathrm{C}$
- The coulomb is a *large* unit: $1\,\mathrm{C}$ contains 6.25×10^{18} elementary charges.
- Typical static-electricity charges are on the order of nanocoulombs (nC) to microcoulombs (μC).

Atoms and Ions

Neutral Atoms

A neutral atom contains **equal numbers of protons and electrons**, giving a net charge of zero.

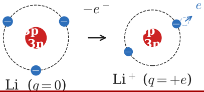
Example: Lithium (Li)

- Neutral Li: 3 protons, 3 electrons $\Rightarrow q = 0$
- Li loses 1 electron $\Rightarrow$ Li^+ ion
- Li^+: 3 protons, 2 electrons $\Rightarrow q = +e$

Definition: Ion

An **ion** is an atom or molecule with a nonzero net charge (unequal numbers of protons and electrons).

- **Cation** (+): lost electrons
- **Anion** (−): gained electrons



Check Your Understanding

Objects contain enormous numbers of protons and electrons, yet most objects appear electrically neutral.

Why don't most objects exhibit obvious static electricity?

Come to lecture to see the answer.

Outline of Part I: Electric Charge

Law of Conservation of Charge

Law of Conservation of Charge

Electric charge is **neither created nor destroyed** in any process.

- Charge can only be *transferred* from one object to another.
- Whenever a positive charge appears somewhere, an equal negative charge simultaneously appears elsewhere.
- The total charge of an isolated system is always constant:

$$Q_{\text{total}} = \text{constant}$$

Example

Rubbing a glass rod with silk: the rod becomes positive, the silk becomes negative.

Check Your Understanding

True or False: rubbing two objects together *creates* electric charge.

Come to lecture to see the answer.

Check Your Understanding

A glass rod is rubbed with silk, leaving the rod with a charge of $+8\,\mathrm{nC}$.

What is the charge on the silk?

Come to lecture to see the answer.

Part II

Conductors, Insulators & Charging

Outline of Part II: Conductors, Insulators & Charging

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Conductors and Insulators

Methods of Charging

Conductors and Insulators

Conductor

A material that **allows charges to move relatively freely** through it.

- Contains **free electrons** not bound to individual atoms.
- Examples: copper, silver, gold, salty water, human body.

Insulator

A material in which **charges cannot move freely** — electrons are tightly bound.

- No free electrons; charge stays put.
- Examples: glass, rubber, dry wood, plastic, pure water.

The mobility of charges in a conductor is approximately 10^{23} times greater than in an insulator — an enormous difference that underlies all of electrical technology.

Check Your Understanding

A negative charge is placed on a metal sphere.

Where does the excess charge reside?

Come to lecture to see the answer.

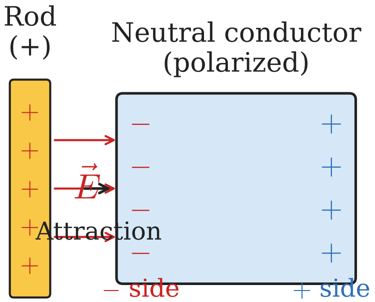
Polarization

Definition: Polarization

Polarization is the separation of positive and negative charges within a neutral object, induced by a nearby external charge. The object remains neutral overall.

Why Neutral Objects are Attracted to Charged Objects

- A charged rod held near a neutral object shifts charge within that object.
- The side *closest* to the rod acquires the opposite sign.
- The attractive force (near side) exceeds the repulsive force (far side).
- Net result: **attraction**, even for a neutral object.



Polar Molecules

Definition: Polar Molecule

A **polar molecule** has a permanent, inherent charge separation — a positive end and a negative end — even though it is electrically neutral overall.

Water: The Most Important Polar Molecule

- H_2O is bent, so its positive (H) and negative (O) sides do not cancel.
- Water's polarity allows it to dissolve salts and carry ions.
- In humid air, water molecules carry ions that neutralize excess surface charge — explaining why static electricity is more common in dry conditions.

Polar molecules attract both positively and negatively charged objects, since either sign of charge can align with the opposite end of the dipole.

Check Your Understanding

A positively charged rod is brought near a small piece of paper (an insulator) without touching it. The paper is attracted to the rod.

Explain why, even though the paper has zero net charge.

Come to lecture to see the answer.

Outline of Part II: Conductors, Insulators & Charging

Three Methods of Charging

1. Charging by Contact (Conduction)

- A charged object touches a neutral conductor; charge transfers by direct contact.
- **Result**: both objects acquire the *same sign* of charge.

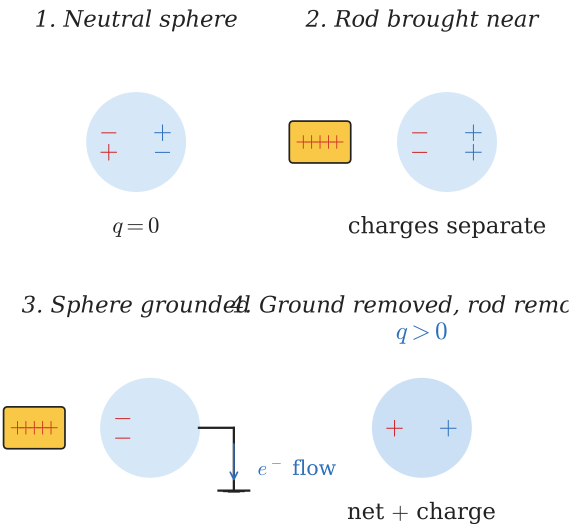
2. Charging by Induction

- A charged object is brought *near* but does not touch a neutral conductor, causing internal charge separation.
- With grounding and then removal of the ground and inducing charge, the conductor acquires the *opposite sign*.

3. Grounding

- Connecting a conductor to Earth allows charge to flow freely to/from Earth's vast reservoir.
- Grounding can neutralize or charge a conductor.

Charging by Induction: Step by Step



Key Result

The conductor ends up with the **opposite sign** of charge from the inducing object, without any contact.

Order matters: the ground must be removed *before* the inducing charge, or the induced charge will flow back.

Check Your Understanding

A negatively charged rod is used to charge a neutral metal sphere by *induction*.

(a) What sign of charge does the sphere acquire?

(b) Did the rod lose any of its charge during this process?

Come to lecture to see the answer.

Check Your Understanding

You touch a positively charged metal rod to a neutral metal sphere (charging by contact).

(a) What sign of charge does the sphere acquire?

(b) After contact, is the rod more positive, less positive, or neutral?

Come to lecture to see the answer.

Part III

Coulomb's Law

Outline of Part III: Coulomb's Law

Coulomb's Law

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Coulomb's Law

Coulomb's Law

Coulomb's Law

The magnitude of the electrostatic force between two point charges q_1 and q_2 separated by distance r:

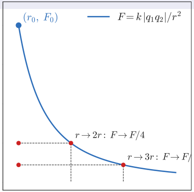
$$\boxed{F = k\frac{|q_1 q_2|}{r^2}}$$

F	Coulomb force (N)
k	$8.99 \times 10^9 \dfrac{\text{N}\cdot\text{m}^2}{\text{C}^2}$
q_1, q_2	point charges (C)
r	separation (m)

- **Attractive** for unlike charges
- **Repulsive** for like charges
- Force is along the line joining the charges
- Newton's Third Law: $\vec{F}_{12} = -\vec{F}_{21}$

Coulomb's Law: The Inverse-Square Law

- $F \propto \frac{1}{r^2}$: the force decreases with the **square** of the distance.
- Double the distance $\Rightarrow$ force becomes $\frac{1}{4}$ as strong.
- Triple the distance $\Rightarrow$ force becomes $\frac{1}{9}$ as strong.



Coulomb's Law and Newton's Law of Gravity share the same mathematical form:

$$F_{\text{Coulomb}} = k\frac{|q_1\,q_2|}{r^2} \qquad F_{\text{gravity}} = G\frac{m_1\,m_2}{r^2}$$

Coulomb Force vs. Gravitational Force

Comparing Forces Between a Proton and Electron

- At atomic separations ($\sim 10^{-10}$ m), the Coulomb force between a proton and electron is approximately $\mathbf{2.27 \times 10^{39}}$ **times stronger** than their gravitational attraction.
- Gravity is negligibly weak at atomic and subatomic scales.
- Electromagnetism governs the structure of atoms, molecules, and all matter.

Despite this, large objects appear electrically neutral because positive and negative charges nearly cancel. Gravity dominates at large scales only because mass is always positive (unlike charge, which can be $\pm$).

Check Your Understanding

Two identical positive charges are separated by a distance r.

If the distance between them is tripled, by what factor does the Coulomb force change?

Come to lecture to see the answer.

Check Your Understanding

Two point charges exert a Coulomb force F on each other.

If the magnitude of each charge is doubled *and* the distance between them is also doubled, what happens to the force?

Come to lecture to see the answer.

Example: Coulomb's Law

Suppose $q_1 = +5\,\mathrm{nC}$ and $q_2 = -5\,\mathrm{nC}$, separated by $r = \sqrt{2} \times 10^{-3}\,\mathrm{m}$.
Find the magnitude of the Coulomb force.

$$F = k\frac{|q_1\,q_2|}{r^2}$$

$$F = \frac{(9.0 \times 10^{9})(5 \times 10^{-9})(5 \times 10^{-9})}{(\sqrt{2} \times 10^{-3})^2}$$

$$F = \frac{(9.0 \times 10^{9})(25 \times 10^{-18})}{2 \times 10^{-6}}$$

$$\boxed{F = 1.13 \times 10^{-1}\,\mathrm{N}}$$

- $1\,\mathrm{nC} = 10^{-9}\,\mathrm{C}$
- $(\sqrt{2} \times 10^{-3})^2 = 2 \times 10^{-6}\,\mathrm{m}^2$
- Opposite charges $\Rightarrow$ attractive force

Check Your Understanding

Charge A exerts a force on charge B.

Does charge B exert a force on charge A?
If so, how does the magnitude compare?

Come to lecture to see the answer.

Part IV

The Electric Field

Outline of Part IV: The Electric Field

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The Electric Field

Electric Field Lines

From Force to Field: Motivation

The Concept of a Field

A **field** maps the force that surrounds any object and acts on another object at a distance — without apparent physical connection.

- **Gravitational field**: created by a mass M; exerts a force on any other mass m:

$$\vec{g} = \frac{\vec{F}}{m} \quad \Rightarrow \quad \vec{F} = m\vec{g}$$

- **Electric field**: created by a charge Q; exerts a force on any other charge q:

$$\vec{E} = \frac{\vec{F}}{q} \quad \Rightarrow \quad \vec{F} = q\vec{E}$$

The field concept lets us describe how distant charges interact without requiring

Definition of the Electric Field

Electric Field $\vec{E}$ — SI Unit: N/C (newtons per coulomb)

The **electric field** at a point is the Coulomb force per unit positive test charge at that point:

$$\boxed{\mathbf{E} = \frac{\mathbf{F}}{q}}$$

$\vec{E}$	electric field (N/C)
$\vec{F}$	force on test charge (N)
q	test charge (C)

- $\vec{E}$ depends only on the *source* charge — not on the test charge q.
- The test charge must be small enough not to disturb the source.

Electric Field from a Point Charge

Electric Field of a Point Source Charge Q

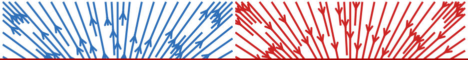
At a distance r from a point charge Q, the electric field magnitude is:

$$\boxed{E = k\frac{|Q|}{r^2}}$$

E	field magnitude (N/C)
k	$8.99 \times 10^9 \, \frac{\mathrm{N{\cdot}m^2}}{\mathrm{C^2}}$
Q	source charge (C)
r	distance from source (m)

Direction of $\vec{E}$

- **Away from** Q if $Q > 0$
- **Toward** Q if $Q < 0$



Force on a Charge in an Electric Field

Any charge q placed in an electric field $\vec{E}$ experiences a force:

$$\boxed{\vec{F} = q\vec{E}}$$

Positive charge in $\vec{E}$

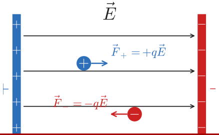
Force is in the **same direction** as $\vec{E}$.

Positive charges follow field lines.

Negative charge in $\vec{E}$

Force is **opposite** to $\vec{E}$, because $q < 0$.

Negative charges move against field lines.



Check Your Understanding

The electric field at a certain point points to the right.

(a) In which direction is the force on a positive charge placed there?

(b) In which direction is the force on a negative charge placed there?

Come to lecture to see the answer.

Check Your Understanding

Why must the test charge used to define the electric field be vanishingly small?

Come to lecture to see the answer.

Check Your Understanding

Two different positive test charges, q_1 and $q_2 = 2q_1$, are placed at the same location near a source charge Q.

(a) Which test charge experiences a greater force?

(b) Which test charge measures a greater electric field at that location?

Come to lecture to see the answer.

Superposition of Electric Fields

Superposition Principle

The total electric field from multiple source charges is the **vector sum** of the fields from each individual charge:

$$\vec{E}_{\text{total}} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \cdots$$

Procedure for Finding $\vec{E}_{\text{total}}$

1. Compute $E_i = k|Q_i|/r_i^2$ for each source charge.
2. Determine the direction of each $\vec{E}_i$ (away from $+$, toward $-$).
3. Decompose into x and y components using geometry/trig.
4. Sum: $E_{\text{total},x} = \sum E_{ix}, \quad E_{\text{total},y} = \sum E_{iy}$.
5. Magnitude: $E_{\text{total}} = \sqrt{E_{\text{total},x}^2 + E_{\text{total},y}^2}$.
6. Direction: $\theta = \tan^{-1}\left(E_{\text{total},y}\right.$

Example: Superposition of Electric Fields

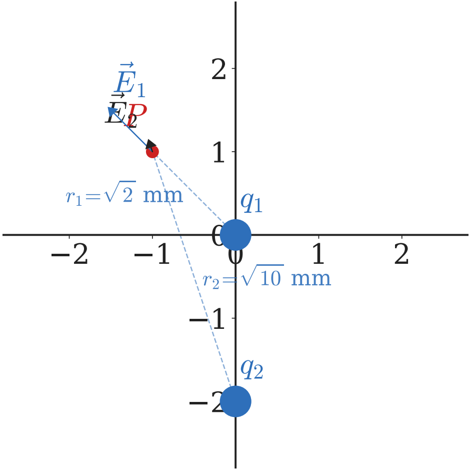
$q_1 = q_2 = 1\,\text{nC}$, fixed at $(0,0)$ and $(0,-2)$ mm. Find $\vec{E}$ at point $P = (-1,1)$ mm.

$$r_1 = \sqrt{1^2 + 1^2} = \sqrt{2}\,\text{mm}$$

$$r_2 = \sqrt{1^2 + 3^2} = \sqrt{10}\,\text{mm}$$

$$E_1 = \frac{(9\times10^9)(1\times10^{-9})}{(\sqrt{2}\times10^{-3})^2} = 4.5\times10^6\ \frac{\text{N}}{\text{C}}$$

$$E_2 = \frac{(9\times10^9)(1\times10^{-9})}{(\sqrt{10}\times10^{-3})^2} = 0.9\times10^6\ \frac{\text{N}}{\text{C}}$$



Example: Superposition (Components)

$$\theta_1 = \tan^{-1}\left(\tfrac{1}{1}\right) = 45^\circ$$

$$E_{1x} = E_1 \cos 45^\circ = 4.5 \times 10^6 \times 0.707 = 3.2 \times 10^6$$

$$E_{1y} = E_1 \sin 45^\circ = 4.5 \times 10^6 \times 0.707 = 3.2 \times 10^6$$

$$\theta_2 = \tan^{-1}\left(\tfrac{3}{1}\right) = 71.5^\circ$$

$$E_{2x} = E_2 \cos 71.5^\circ = 0.9 \times 10^6 \times 0.316 = 0.3 \times 10^6$$

$$E_{2y} = E_2 \sin 71.5^\circ = 0.9 \times 10^6 \times 0.949 = 0.85 \times 10^6$$

Both E_x components point in the $-x$ direction:

$$\begin{aligned} E_{\text{tot},x} &= -(3.2+0.3) \times 10^6 \\ &= -3.5 \times 10^6 \ \tfrac{\text{N}}{\text{C}} \end{aligned}$$

$$\begin{aligned} E_{\text{tot},y} &= (3.2+0.85) \times 10^6 \\ &= 4.05 \times 10^6 \ \tfrac{\text{N}}{\text{C}} \end{aligned}$$

$$E_{\text{tot}} = \sqrt{3.5^2 + 4.05^2} \times 10^6$$

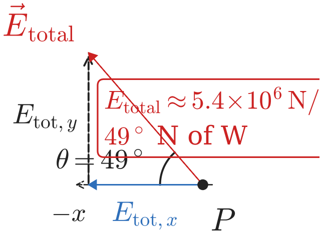
$$\boxed{E_{\text{tot}} \approx 5.4 \times 10^6 \ \tfrac{\text{N}}{\text{C}}}$$

Example: Superposition (Direction)

Direction of $\vec{E}_{\text{total}}$:

$$\theta = \tan^{-1}\left(\frac{E_{\text{tot},y}}{|E_{\text{tot},x}|}\right) = \tan^{-1}\left(\frac{4.05}{3.5}\right) = 49^\circ$$

$$\boxed{\vec{E}_{\text{total}} \approx 5.4 \times 10^6\ \frac{\text{N}}{\text{C}},\ 49^\circ\ \text{N of W}}$$



$E_{\text{tot},x} < 0$ and $E_{\text{tot},y} > 0 \Rightarrow$ field

Outline of Part IV: The Electric Field

The Electric Field

Electric Field Lines

Electric Field Lines

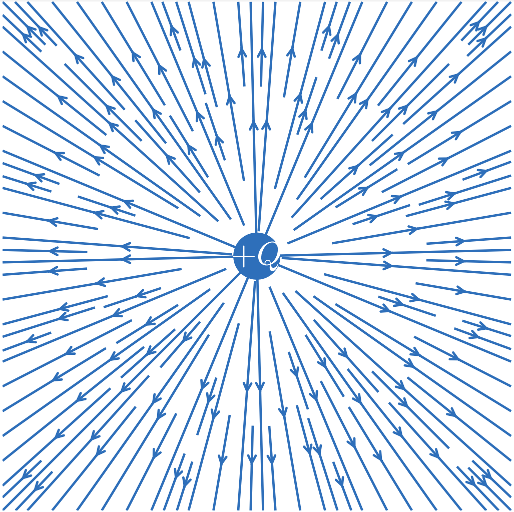
Definition: Electric Field Lines

Electric field lines are a visual map of the direction and strength of an electric field. They are essentially a map of infinitesimal force vectors on a positive test charge.

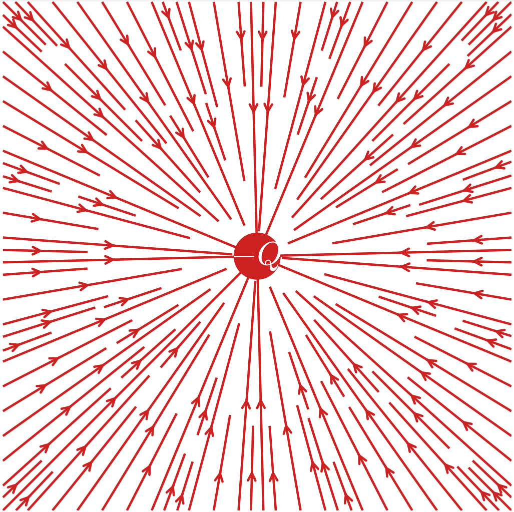
Rules for Drawing Electric Field Lines

- Lines **begin** on positive charges and **end** on negative charges (or extend to infinity for isolated charges).
- The **number of lines** is proportional to the magnitude of the charge.
- The **density of lines** indicates field strength: denser lines = stronger field.
- The **direction of** $\vec{E}$ at any point is tangent to the field line at that point.
- Field lines **never cross** (the field has a unique direction everywhere).

Electric Field Lines: Single Charges

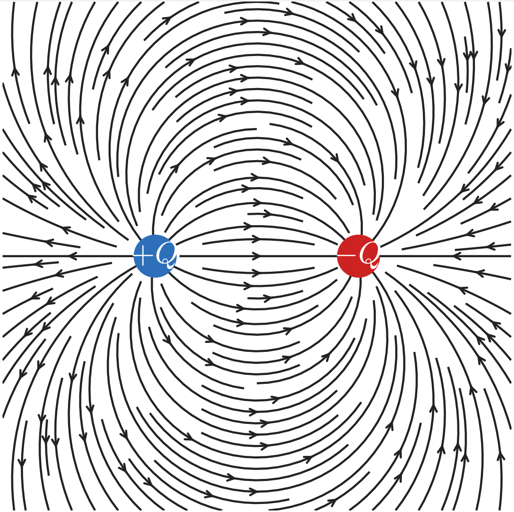


Positive Source Charge

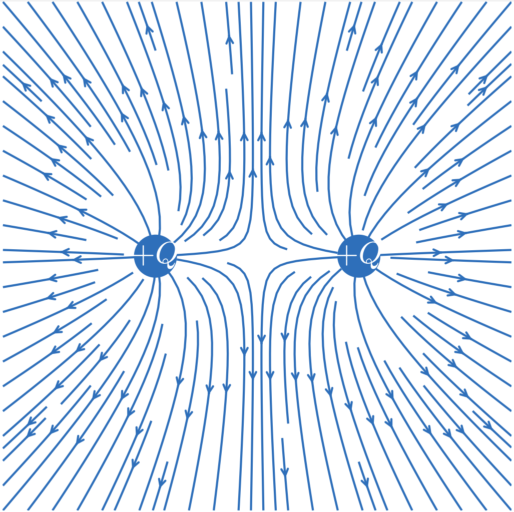


Negative Source Charge

Electric Field Lines: Two Charges



Unlike Charges (+ and −)



Like Charges (+ and +)

Check Your Understanding

Two electric field lines are drawn crossing each other at a point. Is this physically possible? Explain.

Come to lecture to see the answer.

Check Your Understanding

Region A has field lines packed closely together. Region B has field lines spread far apart.

(a) In which region is the electric field stronger?

(b) In which region would a test charge experience a greater force?

Come to lecture to see the answer.

Check Your Understanding

A positive charge $+Q$ and a negative charge $-2Q$ are placed near each other.

If you draw electric field lines for this configuration, which charge should have more lines coming from/going to it?

Come to lecture to see the answer.

Part V

Conductors in Equilibrium & Applications

Outline of Part V: Conductors in Equilibrium & Applications

Outline of Part V: Conductors in Equilibrium & Applications

Conductors in Static Equilibrium

Key Properties

- The electric field **inside** a conductor in static equilibrium is **zero**.
- Excess charge resides entirely on the **outer surface** of the conductor.
- The electric field at the surface is **perpendicular** to the surface.
- Charge (and therefore field) is strongest at **sharp points and corners**.

Reasoning

- If $\vec{E} \neq 0$ inside, free electrons would experience a net force and move — contradicting the assumption of static equilibrium.
- Charges redistribute until the net force on every charge is zero, which requires $\vec{E} = 0$ inside.

The Faraday Cage

Faraday Cage

A **Faraday cage** is a conducting shell that completely shields its interior from external electric fields.

How It Works

- An external field causes charge to redistribute on the outer surface.
- This redistribution creates an internal field that exactly cancels the external field.
- Net field in the interior: zero.

Examples

- Automobiles protect from lightning
- Microwave oven mesh door
- Shielded cables in electronics

Lightning Rods

How a Lightning Rod Works

- Charge accumulates most densely at the *sharp tip* of the rod.
- The intense field at the tip ionizes nearby air, allowing charge to slowly leak off (**corona discharge**).
- This continuously dissipates charge buildup on the building, reducing the likelihood of a sudden lightning strike.
- If lightning does strike, the rod provides a low-resistance path directly to ground.

Invented by Benjamin Franklin (1752), who demonstrated that lightning is an electrical discharge using his famous kite experiment.

Check Your Understanding

A person sits inside a car during a lightning storm.

Are they safe? Explain in terms of the Faraday cage effect.

Would standing next to a tall tree provide similar safety?

Come to lecture to see the answer.

Outline of Part V: Conductors in Equilibrium & Applications

Conductors in Static Equilibrium

Applications of Electrostatics

Applications of Electrostatics

Photocopiers & Laser Printers

- A drum is given a uniform charge.
- A laser selectively discharges areas corresponding to white space.
- Charged toner particles adhere to charged areas.
- Image transfers to paper and is fused with heat.

Electrostatic Air Filters

- Pollutant particles are charged as they enter.
- Oppositely charged plates attract and trap the particles.
- Used in industrial smokestacks and home air purifiers.

Ink-Jet Printers

- Ink droplets are given a precise charge.
- Deflecting plates steer each droplet to the correct position on the page.

Check Your Understanding

Which best describes an *electric field*?

(a) A force acting on a charge.

(b) A region in space where a test charge would experience a force.

(c) The charge on a proton.

Come to lecture to see the answer.

Check Your Understanding

A proton and an electron are placed in a uniform electric field.

(a) Which particle experiences a greater force?

(b) Which particle accelerates more?

Come to lecture to see the answer.